

Maximilian J. Hemmrich

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Education

 **Resident Physician, Otolaryngology-Head&Neck Surgery** 2021-current
UNIVERSITY OF CHICAGO MEDICAL CENTER Chicago, IL

- Expected graduation June 2026
- Passed USMLE Step 3 244 (1/2022)
- Presented Grand Rounds on 3D Printing in OHNS, Optimization of Surgical Scars, and Immunotherapy in Head & Neck Cancer

 **Doctor of Medicine (M.D.)** 2017 - 2021
UNIVERSITY OF CHICAGO PRITZKER SCHOOL OF MEDICINE Chicago, IL

- USMLE Step 1 259 (4/2019) | Step 2 CK 270 (12/2020)
- Court and Sherry Cutting Scholar in Medical Education (2017) | Kappas Trust Fund Scholar (2018)
- Society of Hospital Medicine (SHM) Student Hospitalist Scholar (2018) | Abstract Reviewer for SHM 2020 & 2021 National Conferences
- Pritzker TECH leader - hosted 3-lead EKG building workshop, python coding/statistics course, hosted healthcare hackathon
- Washington Park Free Clinic - performed well-child visits, vaccinations, and sports physicals for uninsured patients
- South Side Science Scholars - taught after-school science lessons on nutrition, physics, and biology to 4th graders on South Side
- Chicago Youth Programs - weekly mentoring sessions to local high school students, assisted with job and college applications

 **Bachelor of Science in Engineering** 2012 - 2016
UNIVERSITY OF MICHIGAN COLLEGE OF ENGINEERING Ann Arbor, MI

- Major in Computer Science & Engineering | Minor in Biochemistry
- GPA 3.90 / 4.00 | Dean's List Fall 2012 - Winter 2016, James B. Angell Scholar
- Instructional Assistant for EECS482 (Operating Systems) and CHEM230 (Physical Chemistry)

Professional Experience

 **Software Engineer - Developer Platforms & Services** 2016-2017
YAHOO! INC Sunnyvale, CA

- Built internal Ruby gem server to reduce dependence on external systems
- Scaled centralized Github Enterprise instance to achieve 99.99% uptime

Research Projects

Oral Premalignancy Progression Prediction | PI: Alexander Pearson, MD, PhD 2024-current
UNIVERSITY OF CHICAGO DEPARTMENT OF HEMATOLOGY AND ONCOLOGY Chicago, IL

- Building model to predict progression of oral premalignant lesions based on deep learning of histopathology
- Combined slides from UChicago, University of Iowa, Brazil, and the NIH to create largest known dataset in the world
- Implemented and validated weakly-supervised learning of epithelial regions of interest
- Training and validation of models using Slideflow ongoing

Oral Cavity Resection Margins | PI: Sameer Alvi, MD 2024-current
ENDEAVOR HEALTH / NORTHSORE UNIVERSITY HEALTH SYSTEM Chicago, IL

- Prospective study to assess impact of resection margins on recurrence rate of cT1 and cT2 oral cavity SCC
- Submitted IRB and created HIPAA-compliant REDCap database
- Patient enrollment currently ongoing

COPD Readmission Prediction | PI: Valerie Press, MD, MPH 2018-2021
UNIVERSITY OF CHICAGO DEPARTMENT OF MEDICINE Chicago, IL

- Built machine learning model using electronic health record data to predict risk of readmission after COPD exacerbation
- Disseminated findings at international, national, and regional conferences
- Received Healthcare Delivery Science Award for research excellence at 2018 Pritzker Summer Research Forum
- Received Health Service Research Abstract Award at 2019 American Thoracic Society International Conference

Head & Neck Cancer Recurrence Database | PI: Alexander Pearson, MD, PhD 2020-2021
UNIVERSITY OF CHICAGO DEPARTMENT OF MEDICINE Chicago, IL

- Built natural language processing classifier to extract tumor data from radiology notes
- Data will be combined with content extracted from pathology notes to build a recurrence database to assess efficacy of treatment modalities and determine predictors of recurrence

- Chart reviews of first 413 COVID patients at UChicago Medical Center
- Literature review for subgroup analysis of COVID in young and obese patients
- Poster accepted at HIV Glasgow 2020 conference

Publications

HEMMRICH, M.J., Auger, S., Chu, V., Blair, E., Portugal, L., Philips, R., Arshad, H., Agrawal, N. (Dec 2024). *Quantifying wake-up events of post-operative head & neck oncology patients on an otolaryngology inpatient floor*. Will submit to American Academy of Otolaryngology-Head and Neck Surgery 2025 Annual Meeting when Call for Science opens Dec 2, 2024

HEMMRICH, M.J., Karcioğlu, A. (Oct 2024). *Parotid oncocytoma recurring as mucoepidermoid carcinoma – a case report*. Submitted to American Head & Neck Society 2025 Annual Meeting at COSM.

Madalone, O., Venigalla, M., HEMMRICH, M.J., Karcioğlu, A. (Oct 2024). *Black hairy tongue as a side effect of immune checkpoint inhibition - a case report*. Submitted to American Head & Neck Society 2025 Annual Meeting at COSM.

HEMMRICH, M.J., Meleca, J., Arshad, H., Agrawal, N. (Aug 2024). *Adverse events associated with the Cook-Swartz Implanatable Doppler Probe for Free Flap Monitoring - a MAUDE Database review*. Submitted to American Head & Neck Society 2025 Annual Meeting at COSM.

Grande, D*, HEMMRICH, M.J*, Agrawal, N. (Oct 2023). *Management of High-Risk Thyroid Cancer*. In: Modi, R., Chaturvedi, P., de Souza, C., eds. *Head and Neck Surgery - Volumes I & II (2nd Edition)*. Forthcoming from Bhalani Medical Book House.

Chokkara, S., Bonomo, M., Kaskovich, S., HEMMRICH, M.J., Carey, K., Venable, L., Rojas, J.C., Churpek, M., Press, V. (Jul 2024). *Comparison of Chart Review and Administrative Data in Developing Predictive Models for Readmissions in Chronic Obstructive Pulmonary Disease*. Submitted to Chronic Obstructive Pulmonary Diseases: Journal of the COPD Foundation.

Rama, N., Wallance, A., HEMMRICH, M.J., Cipriani, N., Pasternak-Wise, O., Agrawal, N., Imbery, T. (Jun 2024). *Skull Base Plexiform Schwannoma of Possible Chorda Tympani Nerve Origin in a Three-year-old Patient*. Published in Otolaryngology Case Reports Vol. 31.

Bonomo, M., Hermesen, M., Kaskovich, S., HEMMRICH, M.J., Rojas, J.C., Carey, K., Venable, L., Churpek, M., Press, V. (Oct 2022). *Using Machine Learning to Predict Likelihood and Cause of Readmission After Hospitalization for Chronic Obstructive Pulmonary Disease Exacerbation*. Published in International Journal of Chronic Obstructive Pulmonary Disease.

Diaz, A., Chavez, J., HEMMRICH, M.J., Smith, H., Donington, J., Portugal, L. (Apr 2022). *Large non-functioning substernal parathyroid cyst: A case report and review of the literature*. Published in International Journal of Surgery Case Reports.

Oral Presentations

HEMMRICH, M.J., Churpek, M., Press, V. (Sep 2019). *Derivation and Validation of a COPD Readmission Risk Prediction Tool*. Presented by MJH at Society of General Internal Medicine Regional Meeting, Minneapolis, MN.

HEMMRICH, M.J., Churpek, M., Press, V. (Mar 2019). *Derivation and Validation of a COPD Readmission Risk Prediction Tool*. Presented by MJH at Society of Hospital Medicine National Conference, National Harbor, MD.

HEMMRICH, M.J., Churpek, M., Press, V. (Aug 2018). *Derivation of a COPD Readmission Risk Prediction Tool*. Presented by MJH at University of Chicago Pritzker Summer Research Forum, Chicago, IL.

Poster Presentations

Hermesen, M., Bonomo, M., HEMMRICH, M.J., Kaskovich, S., Churpek, M., Press, V. (Aug 2020). *Predicting 90-Day Readmission Risk Among Chronic Obstructive Pulmonary Disease Patients: Gold Standard Cohort vs. Administrative Data*. Presented by MH at American Thoracic Society International Conference, Virtual due to COVID.

Bonomo, M., Hermesen, M., Kaskovich, S., HEMMRICH, M.J., Churpek, M., Press, V. (Aug 2020). *Toward Personalized Care: The Value of Chart Reviewed Datasets in Predicting Cause of Readmission*. Presented by MB at American Thoracic Society International Conference, Virtual due to COVID.

Sherer, R., UChicago COVID ID Consult Service Study Group (Sep 2020). *Cancer, transplantation, and other immunocompromising conditions were not significantly associated with severe COVID-19 or death in hospitalized COVID-19 patients in Chicago*. Presented by RS at HIV Glasgow Congress, Virtual due to COVID.

HEMMRICH, M.J., Kaskovich, S., Churpek, M., Press, V. (May 2019). *Accuracy Comparison of a Machine Learning Readmission Prediction Model with HOSPITAL and PEARL Scores for COPD Inpatients*. Presented by MJH at American Thoracic Society International Conference, Dallas, TX.

Kaskovich, S., HEMMRICH, M.J., Churpek, M., Press, V. (May 2019). *Matching Patients with COPD to Personalized Care: A Novel Machine-Learning Tool to Predict Cause of 90-day Readmission*. Presented by SK at American Thoracic Society International Conference, Dallas, TX.

Skills

Languages English (native) | German (native)

Tech Stack Python (Pandas/Numpy/Jupyter) | C++ | Ruby | Git

Personal Interests Tennis | Weightlifting | Chess | Investing